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THE TITANLIFT

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The Titan Lift

Institute: Cambodia Academy of Digital Technology

Course: Algorithm & Computation Thinking

Lecturer: Han Leangsiv

Group: 5

Team: 9

Date: 3rd July 2026

Presented by:



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Oeurn Meylih

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Introduction & Overview

- TitanLift is a smart elevator system in 10-floor building where floor 9 is a VIP and 10 is a sky bar
- TitanLift got inspired by THE TITANIC which had 4 elevators, 3 reserved for first class and 1 for everyone else
- TitanLift got from "TITAN + LIFT" which TITAN means strength, scale and power ,with "LIFT" simply the elevator itself, the thing that carries us up and down



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Key Features of System

01 Floor Request Queue(Priority-Based)

02 VIP Express Mode

03 Card Scan Authentication

04 Emergency Alarm Mode

05 Passenger Capacity Limit

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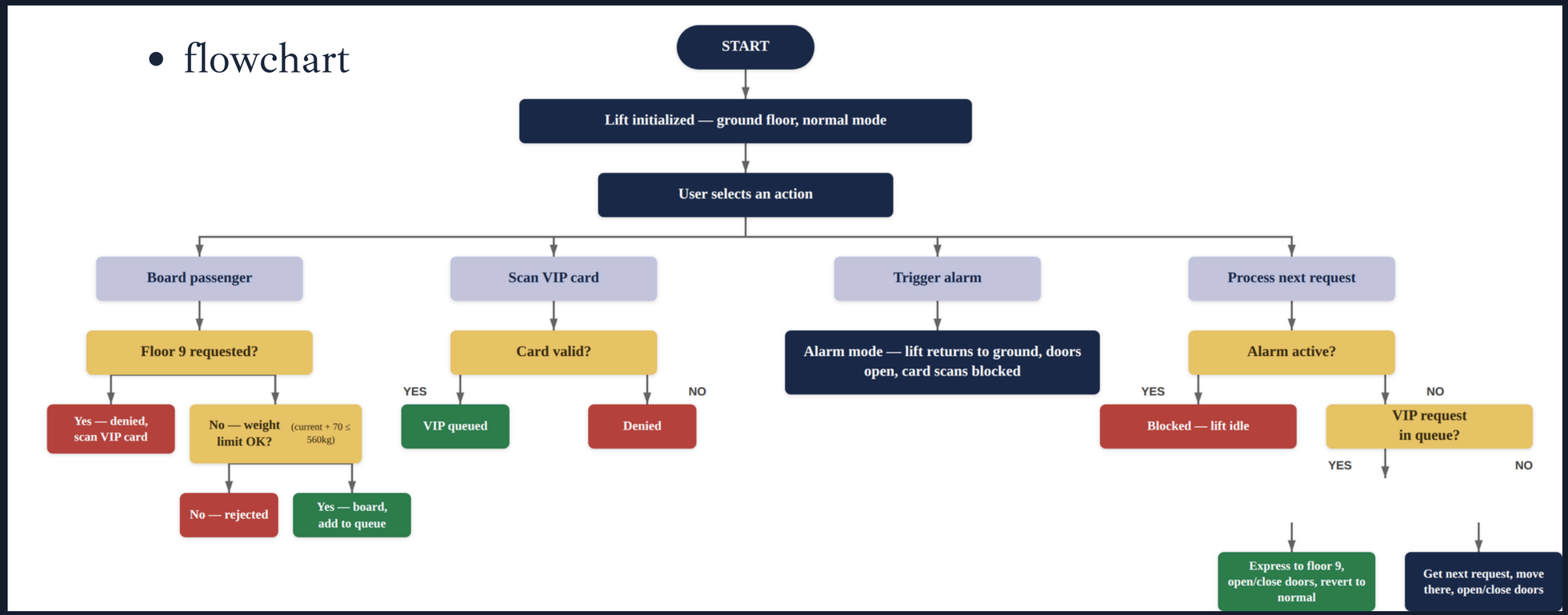
Key Features of System

	Features	Description
01	Floor Request Queue	Stores passengers, requests, processes highest priority first
02	VIP Express	moves directly to floor 9
03	Card Scan Authentication	scan VIP mode , VIP card IDs
04	Emergency Alarm Mode	the alarm requests the cabin to stop the current request
05	Passenger Capacity Limit	rejects entry when full of 8

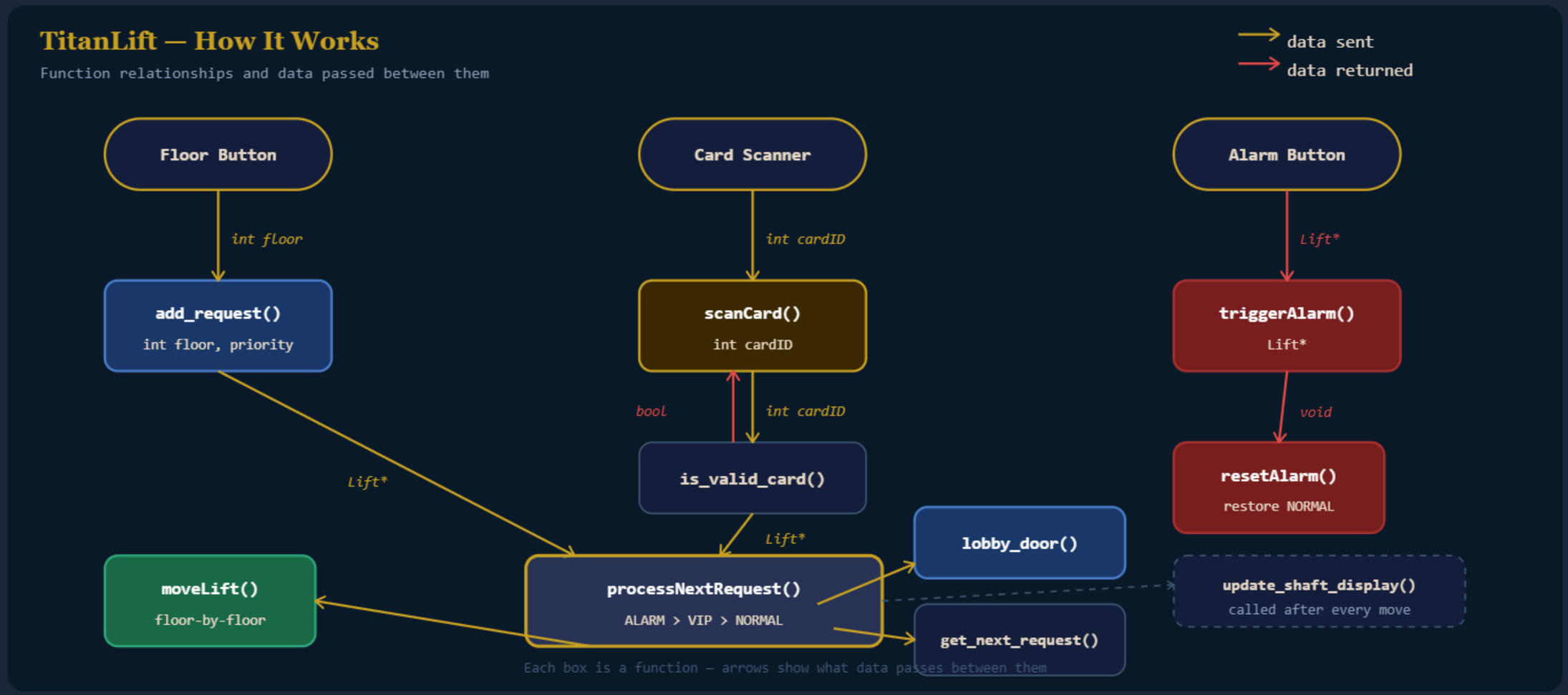
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How It Works

- flowchart



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KEY ACHIEVEMENTS OF OUR SYSTEM

01 VIP Express Override work end to end

02 Priority Queue handles mixed requests

03 Emergency alarm response

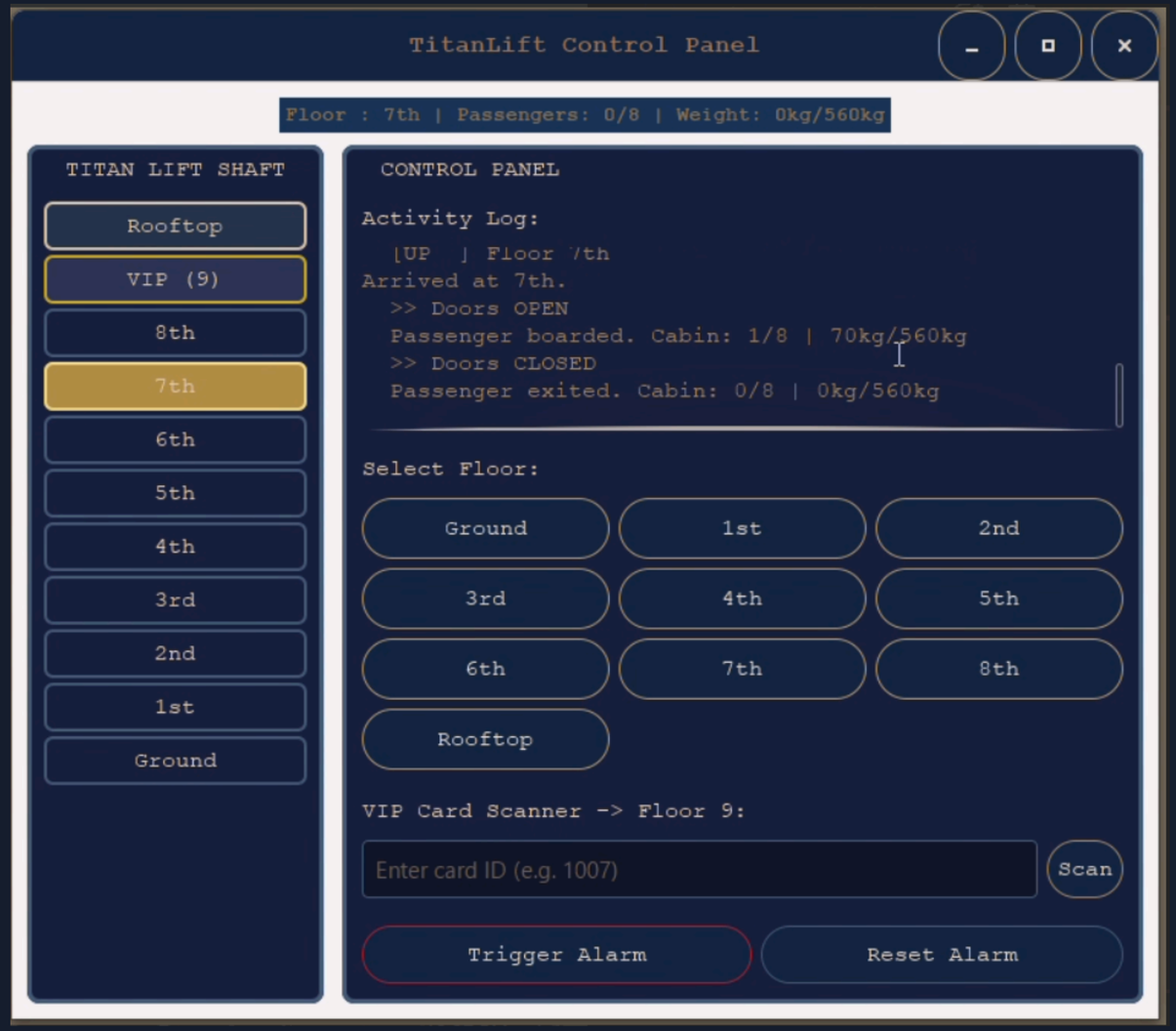
04 Secure card-based access control

05 Passenger capacity enforcement

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RESULT

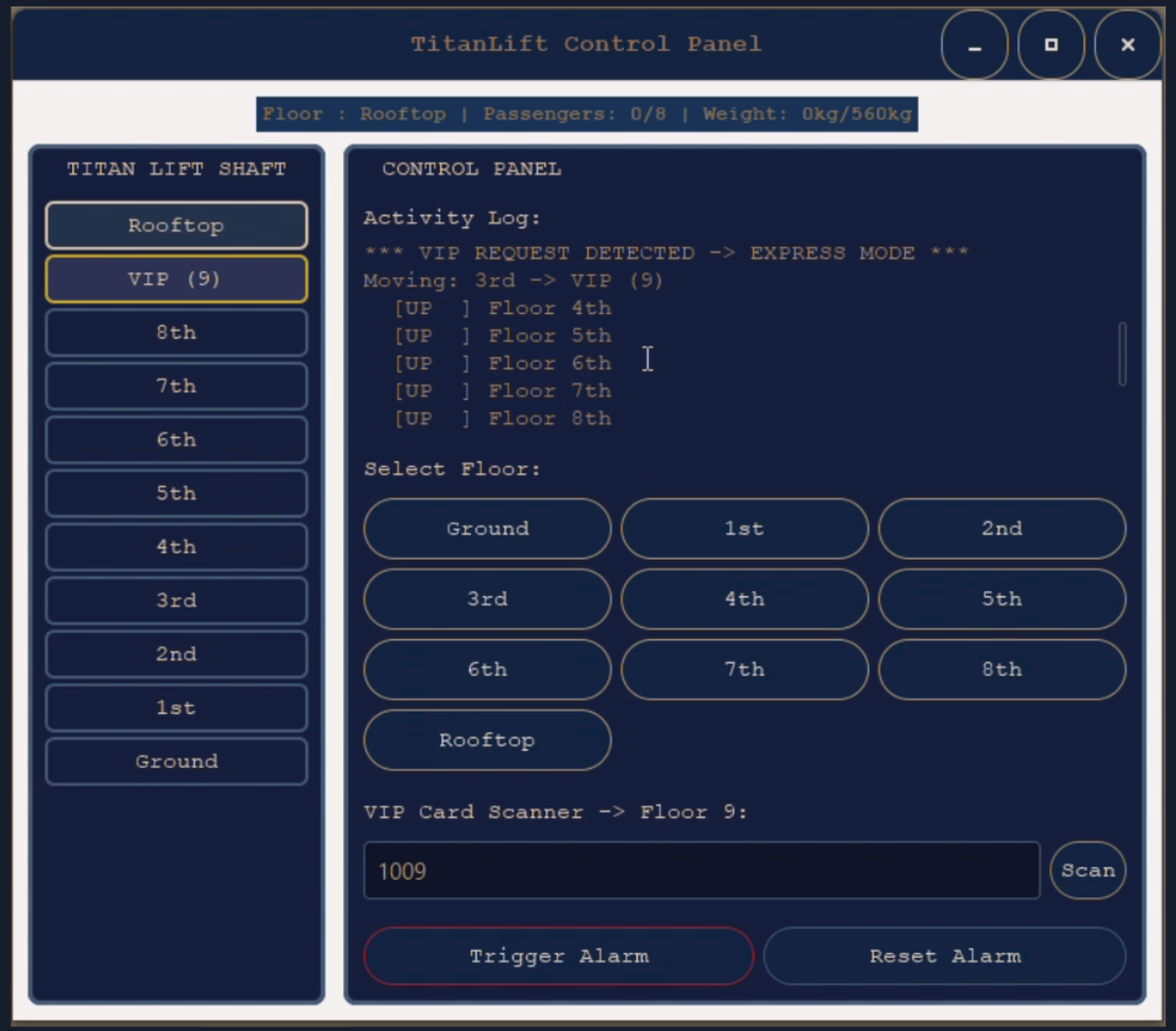




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RESULT

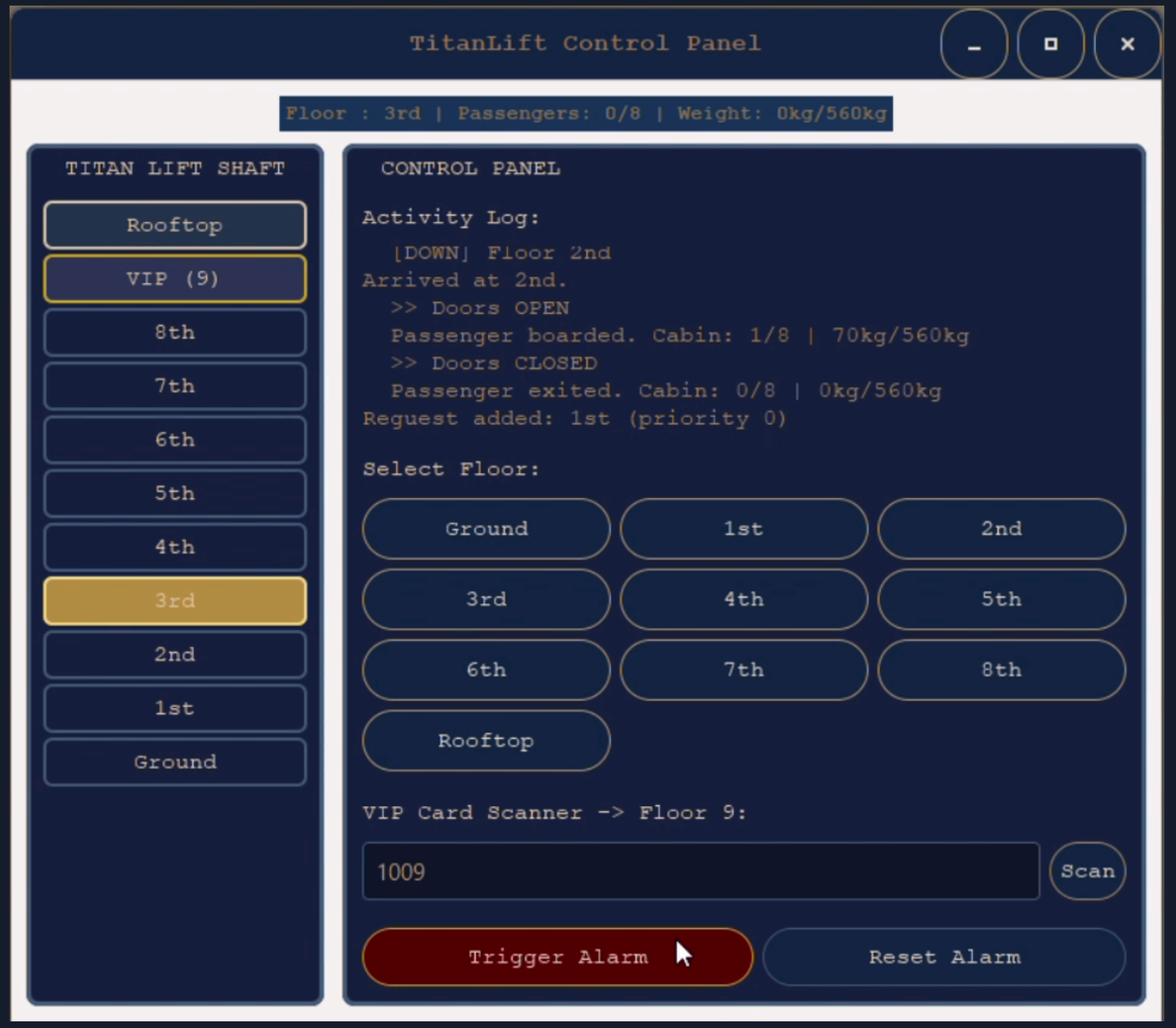




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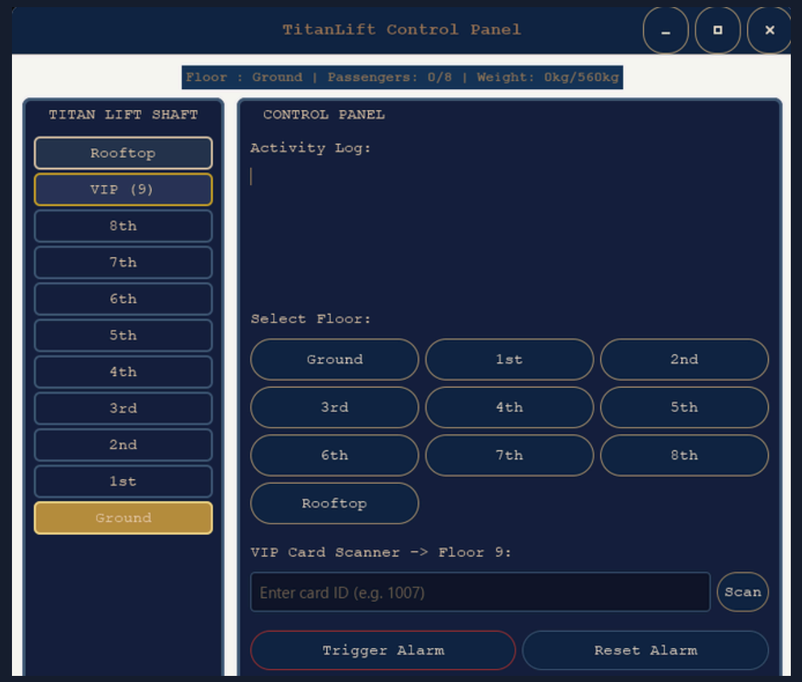
RESULT





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DEMONSTRATION



NOW
LET'S GIVE
A TRY

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Conclusion

- **Summary:**

TitanLift started from a problem everyone’s quietly felt nowsday, waiting in an elevator that stops at every floor when you’re already running late. We took the Titanic’s old idea of priority access and rebuilt it as something fair and temporary: a verified VIP card scan triggers Express Mode, sending the elevator straight to Floor 9, then automatically returning to normal service once that trip is done.

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Challenges/Difficulty



- We face many problems about coding
- Our coding isn't good enough to build the whole things
- Never know about GUI
- Start doing GUI from Zero
- Spent almost 3days just to install GUI and we use GTK

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Lesson Learnt

- Improved our coding skills
- Understand well how to use struct, function and string
- Know how to sovle the real-problem
- Starting to know about GUI
- Learnt how to work as a team



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By Group5 Team9

Thank You